Cover Page

Mobile Forensics (MBF 712S) Lab Investigation	
Lab No:	*Lab 4*
Lead Investigator:	*Titus Shaangelo 222084774* *Barnabas Iihuhwa 223081701* *Tjijenda.u 220113009*
Qualifications:	*B.COMP.SC. IN CYBER SECURITY*
Nature of Case: **(Lab Objective)**	*The objective of this lab was to extract, analyze, and interpret SIM card data using a non-forensically sound tool. The lab emphasizes understanding the limitations of such tools while applying digital forensic investigation principles.*
Due Date:	*Monday, 20 April 2026*

Description of Evidence		**Vendor Name**	**Model /Serial No.**
Item #1	Phonebook_Sebastian.phn	Dekart Software	SIM Manager
Item #2	Phonebook_Nau.phn	Dekart Software	SIM Manager
Item #3	Printing messages to PDF	Do PDF	Software
Item #4	SIM card reader for file access	Computer/Laptop	N/A

1. Introduction

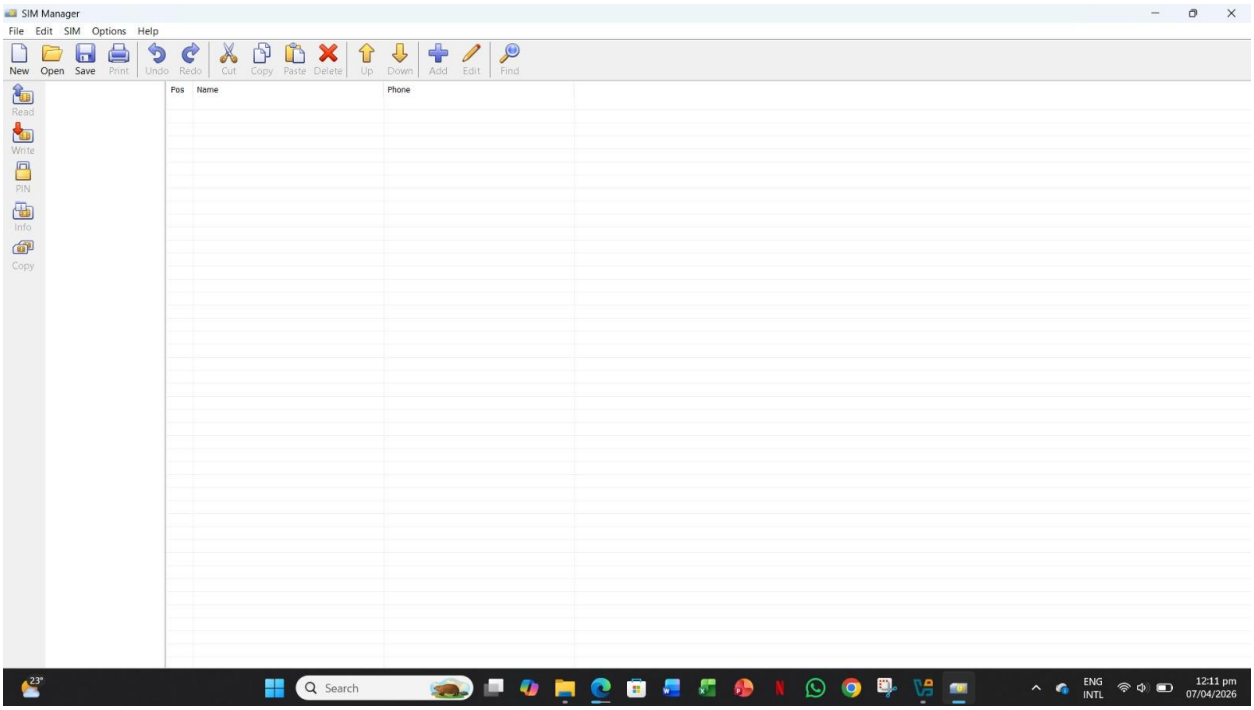
In this lab, we explored how to extract and examine SIM card data using Dekart SIM Manager**,** which is not forensically sound. SIM cards store contacts, messages, and network information that may be relevant in investigations. The lab demonstrates how data can be extracted, analyzed, and timelines created, while highlighting the risks of using tools that may alter evidence.

2. Lab Procedures

1, Install and Open SIM Manager

- Installed SIM Manager on the computer.
- Clicked OK on the unregistered copy warning to continue.

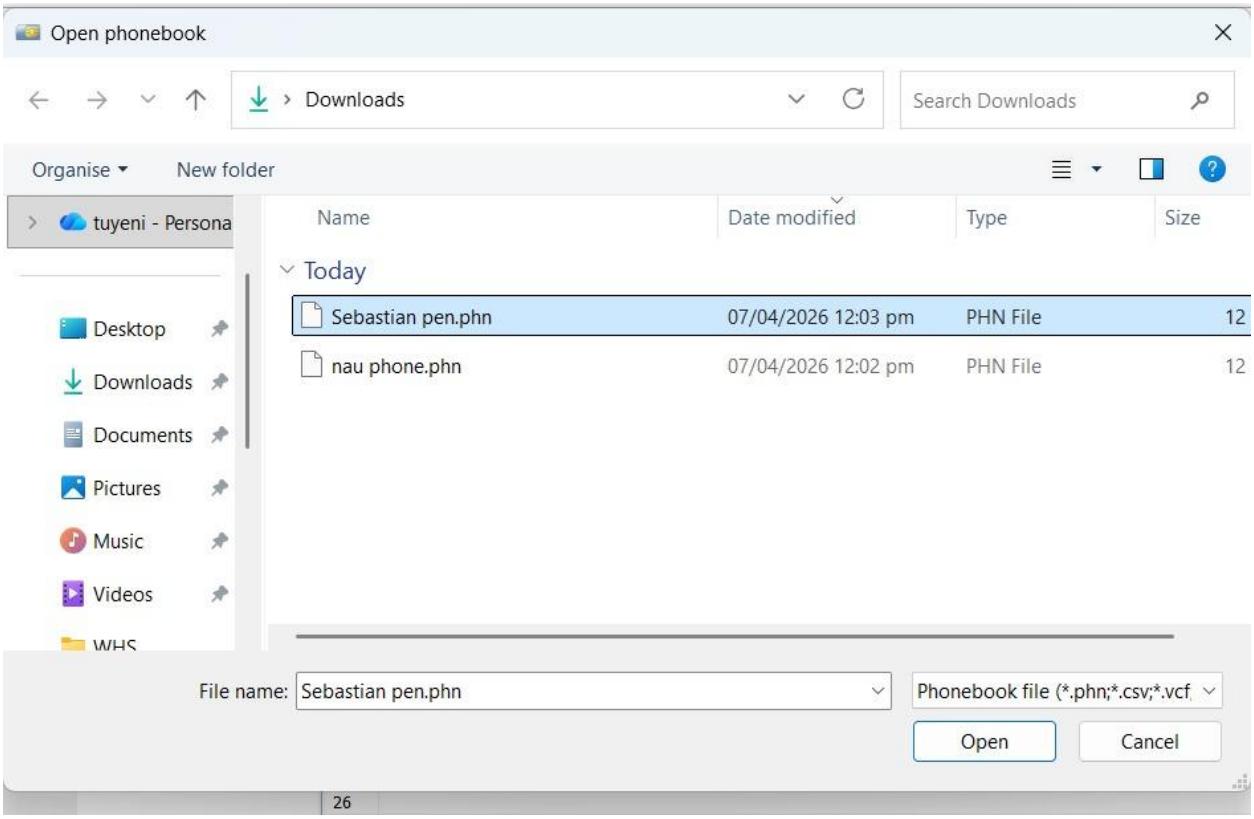
Screenshot: After opening SIM Manager



2, Open Sebastian's SIM File

- Clicked the Open toolbar icon.
- Navigated to the work folder and selected Phonebook_Sebastian.phn.
- Clicked OK to load the file.

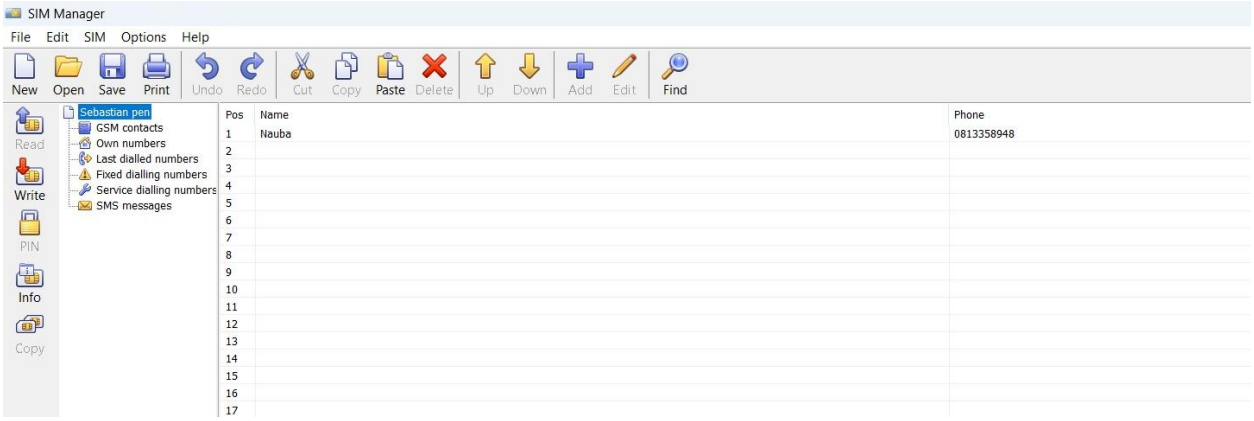
Screenshot: Showing file loaded.



3, View Contact Details

- Selected Phonebook_Sebastian from the left panel.
- Observed Sebastian's full name and phone number on the right panel.

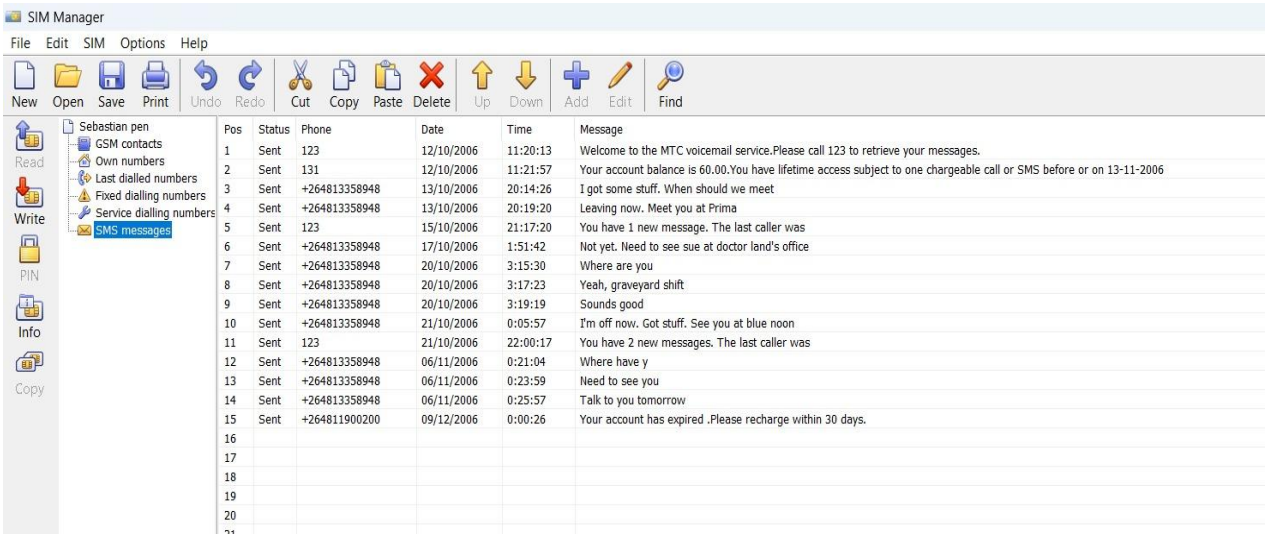
Screenshot: Showing Sebastian's contact details.



4, Examine SMS Messages

- Clicked SMS Messages icon on the left.
- Reviewed all messages displayed in the right panel.

Screenshot: Showing example messages.

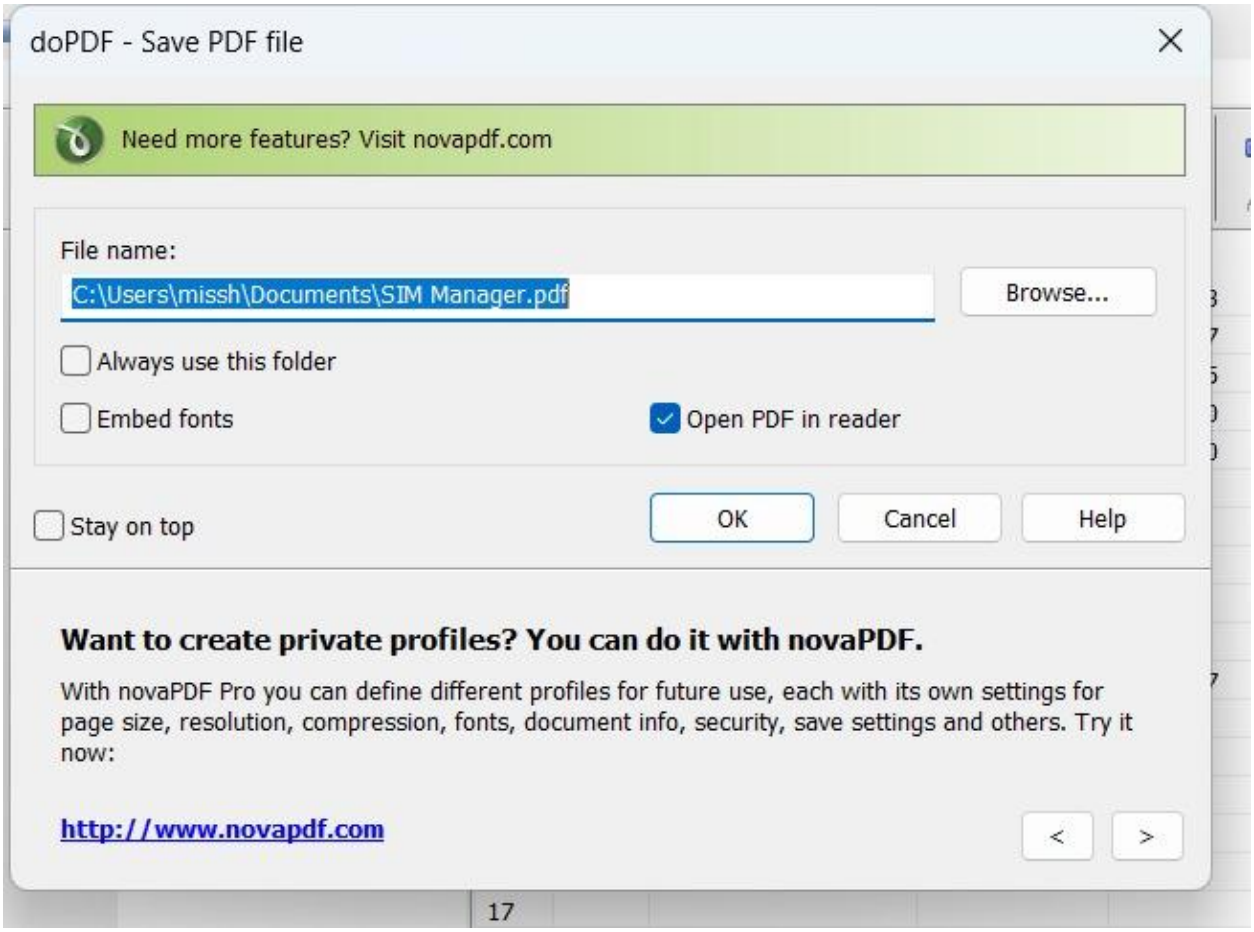


Pos	Status	Phone	Date	Time	Message
1	Sent	123	12/10/2006	11:20:13	Welcome to the MTC voicemail service.Please call 123 to retrieve your messages.
2	Sent	131	12/10/2006	11:21:57	Your account balance is 60.00.You have lifetime access subject to one chargeable call or SMS before or on 13-11-2006
3	Sent	+264813358948	13/10/2006	20:14:26	I got some stuff. When should we meet
4	Sent	+264813358948	13/10/2006	20:19:20	Leaving now. Meet you at Prima
5	Sent	123	15/10/2006	21:17:20	You have 1 new message. The last caller was
6	Sent	+264813358948	17/10/2006	1:51:42	Not yet. Need to see sue at doctor land's office
7	Sent	+264813358948	20/10/2006	3:15:30	Where are you
8	Sent	+264813358948	20/10/2006	3:17:23	Yeah, graveyard shift
9	Sent	+264813358948	20/10/2006	3:19:19	Sounds good
10	Sent	+264813358948	21/10/2006	0:05:57	I'm off now. Got stuff. See you at blue noon
11	Sent	123	21/10/2006	22:00:17	You have 2 new messages. The last caller was
12	Sent	+264813358948	06/11/2006	0:21:04	Where have y
13	Sent	+264813358948	06/11/2006	0:23:59	Need to see you
14	Sent	+264813358948	06/11/2006	0:25:57	Talk to you tomorrow
15	Sent	+264811900200	09/12/2006	0:00:26	Your account has expired .Please recharge within 30 days.
16					
17					
18					
19					
20					

5, Print Messages to PDF

- Clicked Print toolbar icon.
- Accepted default selections.
- Printed messages to PDF using Do PDF.

Screenshot: Showing print-to-PDF confirmation.


doPDF - Save PDF file
Need more features? Visit novapdf.com
File name:
C:\Users\missh\Documents\SIM Manager.pdf
Browse...
Always use this folder
Embed fonts
Open PDF in reader
Stay on top
OK
Cancel
Help
Want to create private profiles? You can do it with novaPDF.
With novaPDF Pro you can define different profiles for future use, each with its own settings for page size, resolution, compression, fonts, document info, security, save settings and others. Try it now:
http://www.novapdf.com
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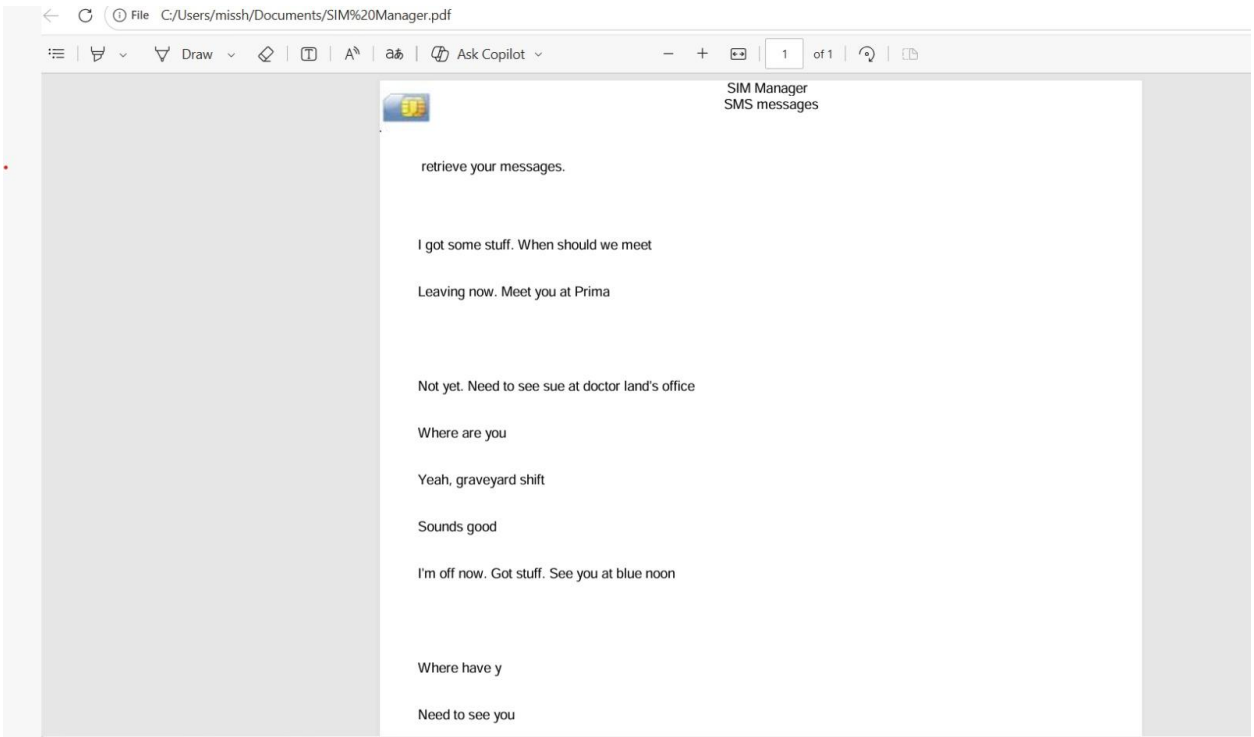

File C:/Users/missh/Documents/SIM%20Manager.pdf
Draw
Ask Copilot
1 of 1
SIM Manager
SMS messages
retrieve your messages.
I got some stuff. When should we meet
Leaving now. Meet you at Prima
Not yet. Need to see sue at doctor land's office
Where are you
Yeah, graveyard shift
Sounds good
I'm off now. Got stuff. See you at blue noon
Where have y
Need to see you

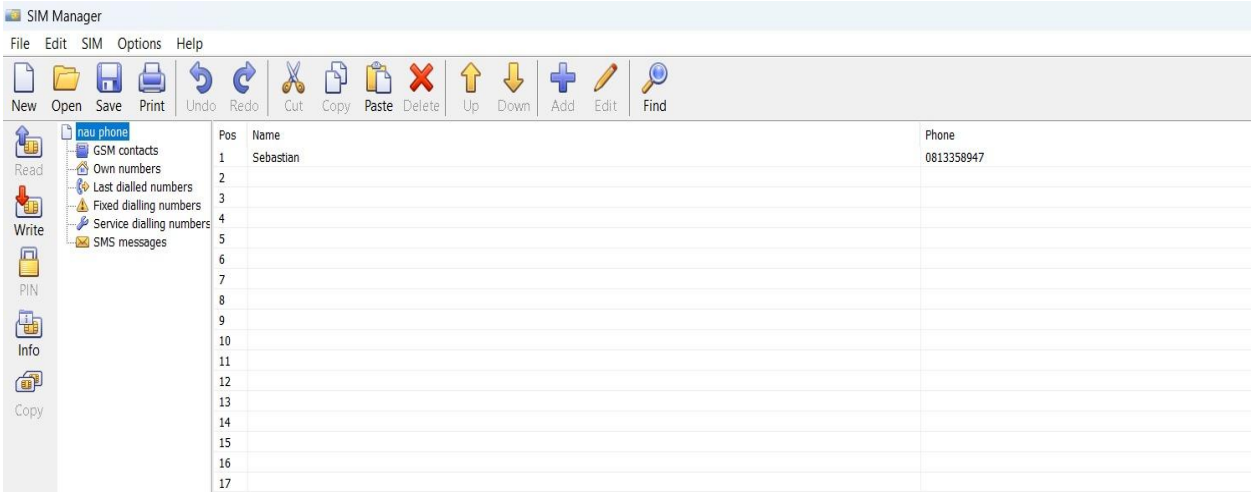
6, Analyze SIM Tool Limitations

- Explored menu items.
- Noted that SIM Manager allows modifying SIM data, making it non-forensically sound.
- Closed Sebastian's file via File → Close.

7, Open Nau's SIM File

- Clicked Open toolbar icon.
- Selected Phonebook_Nau.phn and clicked OK.

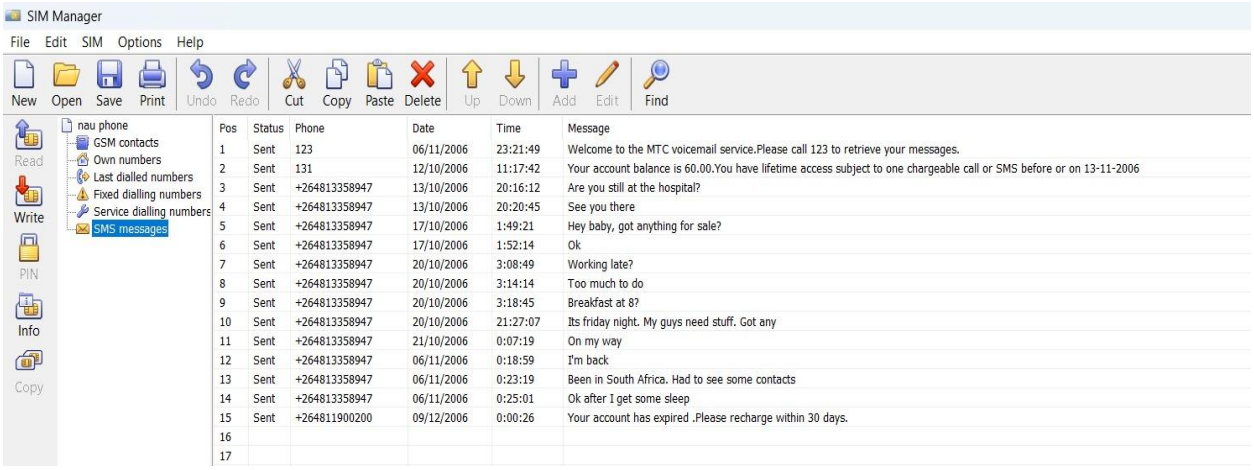
  Screenshot: Showing Nau's SIM loaded.



8, Determine Full Name & SMS Analysis

- Found Nau's full first name in contact details.
- Clicked SMS Messages icon to review messages.
- Observed two different SMS centers, likely indicating different networks or message routing.

  Screenshot: Showing SMS centers.

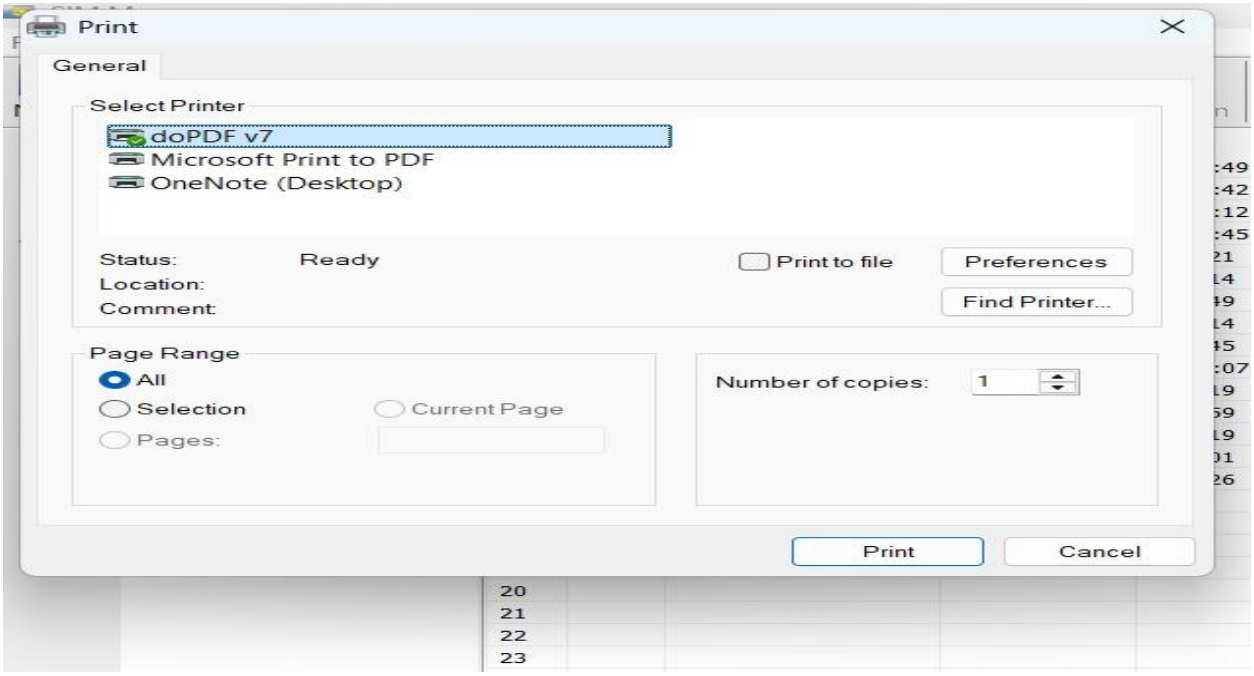


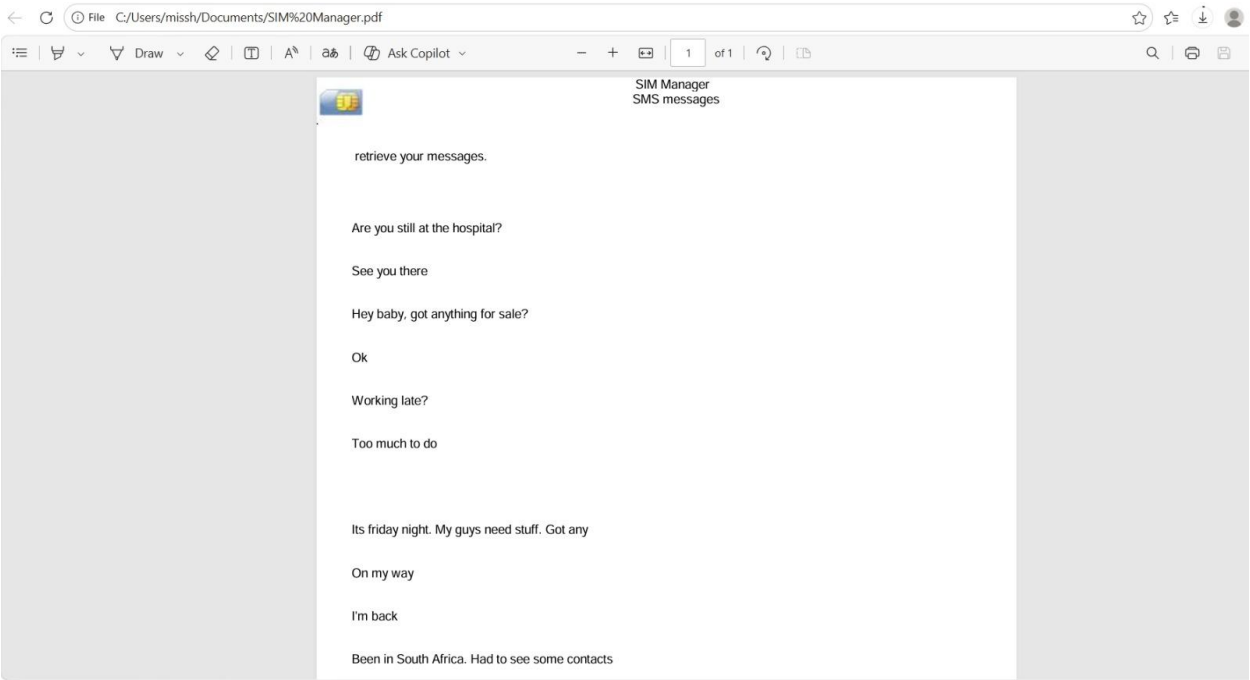
Pos	Status	Phone	Date	Time	Message
1	Sent	123	06/11/2006	23:21:49	Welcome to the MTC voicemail service.Please call 123 to retrieve your messages.
2	Sent	131	12/10/2006	11:17:42	Your account balance is 60.00.You have lifetime access subject to one chargeable call or SMS before or on 13-11-2006
3	Sent	+264813358947	13/10/2006	20:16:12	Are you still at the hospital?
4	Sent	+264813358947	13/10/2006	20:20:45	See you there
5	Sent	+264813358947	17/10/2006	1:49:21	Hey baby, got anything for sale?
6	Sent	+264813358947	17/10/2006	1:52:14	Ok
7	Sent	+264813358947	20/10/2006	3:08:49	Working late?
8	Sent	+264813358947	20/10/2006	3:14:14	Too much to do
9	Sent	+264813358947	20/10/2006	3:18:45	Breakfast at 8?
10	Sent	+264813358947	20/10/2006	21:27:07	Its friday night. My guys need stuff. Got any
11	Sent	+264813358947	21/10/2006	0:07:19	On my way
12	Sent	+264813358947	06/11/2006	0:18:59	I'm back
13	Sent	+264813358947	06/11/2006	0:23:19	Been in South Africa. Had to see some contacts
14	Sent	+264813358947	06/11/2006	0:25:01	Ok after I get some sleep
15	Sent	+264811900200	09/12/2006	0:00:26	Your account has expired .Please recharge within 30 days.
16					
17					

9, Print Nau's Messages to PDF

- Followed same printing procedure as Step 3.5.

Screenshot: Messages printed to PDF.





SIM Manager
SMS messages

retrieve your messages.

Are you still at the hospital?

See you there

Hey baby, got anything for sale?

Ok

Working late?

Too much to do

Its friday night. My guys need stuff. Got any

On my way

I'm back

Been in South Africa. Had to see some contacts

10, Compare Messages and Create Timeline

- Compared messages from Sebastian and Nau.
- Correlated timestamps to create a chronological timeline of communication.

Sample Timeline:

Date	Time	Sender	Receiver	Message
12/10/2006	11:17–11:21	Sebastian → Own numbers	N/A	Welcome to MTC voicemail; account info
13/10/2006	20:14	Sebastian → +264813358948	Nau	Initial contact message

Date	Time	Sender	Receiver	Message
13/10/2006	20:16–20:20	Nau → +264813358947	Sebastian	Confirming initial contact / service
17/10/2006	01:49	Nau → +264813358947	Sebastian	First personal message: "Ok"
17/10/2006	01:51	Sebastian → +264813358948	Nau	"Not yet. Need to see Sue at doctor's office"
20/10/2006	03:08–03:19	Nau → +264813358947	Sebastian	Multiple short updates: "Too much to do", "Yeah, graveyard shift", "Sounds good"
20/10/2006	03:15–03:17	Sebastian → +264813358948	Nau	Checking status: "Where are you?"
20/10/2006	21:27	Nau → +264813358947	Sebastian	"On my way"
20/10/2006	03:15–03:19	Sebastian → +264813358948	Nau	"Where are you?" / "Yeah, graveyard shift"
21/10/2006	00:05	Sebastian → +264813358948	Nau	"I'm off now. Got stuff. See you at Blue Noon"
21/10/2006	22:00	Sebastian → 123	N/A	New voicemail message notification
06/11/2006	00:07–00:25	Nau → +264813358947	Sebastian	Updates and casual messages: "Been in South Africa. Had to see some

Date	Time	Sender	Receiver	Message
				contacts", "Ok after I get some sleep"
06/11/2006	00:21–00:25	Sebastian → +264813358948	Nau	"Where have y...", "Need to see you", "Talk to you tomorrow"
09/12/2006	00:00	Both → +264811900200	N/A	Account expiration notice: "Your account has expired. Please recharge within 30 days."

3. **Lab Challenges/Problems Encountered**
   NONE – The lab was completed successfully.

4. **Conclusion**

   This lab demonstrated that while non-forensically sound tools such as Dekart SIM Manager can effectively extract and display SIM card data, they are inherently limited in forensic investigations due to their potential to alter or overwrite evidence, which compromises data integrity. The analysis of contacts, SMS messages, and the creation of a communication timeline highlighted important patterns in the data, including differences in SMS centers that may indicate multiple service providers or message routing paths. Overall, the exercise reinforced the critical need for using forensically sound tools in formal investigations, while also illustrating the value of careful analysis and documentation in understanding mobile communication data. Further verification and the use of validated forensic software would be required before any conclusions could be considered legally reliable.

5. **The role played by each group member:**

Member	Role
Tjijenda.u 220113009	Coordinated the lab, oversaw procedures, ensured all steps were completed correctly.
Barnabas Iihuhwa 223081701	Extracted SIM data, captured screenshots, and assisted with tool usage.
Titus Shaangelo 222084774	Analyzed messages, created timeline, wrote the report in own words, and verified accuracy.

References

1. Casey, E. (2011). *Digital Evidence and Computer Crime: Forensic Science, Computers and the Internet.* 3rd Edition. Academic Press.
2. Dekart Software. (2026). *SIM Manager User Guide.* Retrieved from https://www.dekart.com
3. Nelson, B., Phillips, A., & Steuart, C. (2015). *Guide to Computer Forensics and Investigations.* 5th Edition. Cengage Learning.